4. GAMMA- RAY SPECTROSCOPY WITH Ge(Li) AND ABUNDANCE OF URANIUM ISOTOPES

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ABSTRACT

The objective of the experiment is to understand the isotopic abundance of Uranium and characteristics of measured gamma ray spectra. A high purity germanium detector drifted with Lithium HPGe(Li) is used to measure gamma rays in high resolution. This is called gamma-ray spectroscopy.

The two isotopes of Uranium considered here are Uranium-235 and Uranium-238. Prime objective is to find the isotopic abundance of Uranium-235 and recognize if the sample is natural, enriched or depleted. Initially the data spectrum was taken for Cesium-137, Barium-133, and Cobalt-60. Energy calibration formula was obtained using these sources and its accuracy is measured using the Europium-152 data values. The relative detector efficiency was calculated using the obtained data to identify the detector efficiency for the Uranium peaks. The detector efficiency is needed to calculate the isotopic abundance of Uranium-235.

According to our data, we obtained the isotopic abundance ratio as 0.0072. It would mean that the Uranium used in our experiment is natural.

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1. BACKGROUND AND INTRODUCTION

The objective is to understand the gamma-ray emissions of the Uranium isotopes (U235 and U238) and to calculate the isotopic abundance ratio. It was also important to understand how to work with the Germanium detector and analyze the energy spectrum received to obtain the energy calibration formula and then to get the isotopic abundance ratio.

Initially using the germanium detector, the energy spectrum was obtained for Cobalt-60, Cesium-137 and Barium-133. Data collected using these sources were used to obtain the energy calibration formula. The accuracy of the calibration formula was determined by cross-checking the values obtain using the fit function to the corresponding gamma energies of the Europium-154 data. The next task is to obtain the detector efficiency. The number of counts and the respective gamma ray intensities were used to calculate the efficiency of the Europium peaks. Efficiency vs the gamma energy values graph was used to get the efficiency of the Uranium peaks. Once we knew the efficiency and the number of counts for Uranium, the isotopic abundance ratio was calculated

which was used to determine whether the uranium-235 sample used was natural depleted or enriched.

Uranium-238: Most common in nature. This isotope has no possibility of sustaining a chain reaction in a thermal-nuclear reactor, hence non-fissile. But it is fissionable due to fast neutrons. It has a half-life of 1.41*10^17 seconds. (International Atomic Energy Agency) As a result of its natural abundance and half-life relative to other radioactive elements it produces an approximately 40% of radioactive heat within earth.

Uranium-235: Fissile. The isotope can endure a nuclear chain reaction. Half-life of 703.8 million years. (International Atomic Energy Agency) Highly enriched Uranium-235 are used in the making of nuclear weapons.

Uranium enrichment is where the proportion of Uranium-235 is increased in the natural uranium-ore. Natural uranium contains a higher proportion of Uranium-238 and does not undergo a nuclear chain reaction. Enriched uranium is used as nuclear fuel in nuclear reactors. (Uranium Enrichment, 2020) There are several enrichment methods such as Gas diffusion, Gas centrifugation and Laser enrichment.

Once the enriched uranium is removed, the by-product uranium mixture has a reduced concentration of U235 and U234. This by-product is known as depleted uranium and its percentage fraction by weight is less than 0.711%. (International Atomic Energy Agency) Depleted uranium is less radioactive as it has less U234 and U235 per unit mass and all traces of decay products beyond U234 and Th231 are removed during extraction and chemical processing prior to enrichment.

The abundance of different isotopes is of vital importance in various science and nuclear fields. A higher concentration of uranium-235 isotope is used for nuclear fuel in a nuclear reactor. When the uranium-235 is enriched, it is fissionable in light water reactors. The alpha decay chain of uranium-238 is known as the uranium series while the alpha decay of uranium-235 is known as the actinium series. The decay of the different isotopes leads to formation of a different set of elements.

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2. THEORY

Isotopes are atoms that has the identical atomic number but distinct mass numbers. There are isotopes that are naturally occurring as well as artificially made. In a given element the isotopes do not exist in equal ratios. For an example Uranium consists of 99.3% uranium-238, 0.7% uranium-235 and 0.01% uranium-234.

Isotopic abundance is important in the detection and identification of presence of uranium isotopes based on their gamma ray emissions in gamma ray spectroscopy.

Radioactive decay is the process where energy is emitted as ionizing radiation. Alpha, beta and gamma particles are the ionizing radiation that is emitted. Elements can be stable and unstable. Stable elements can be found in nature. Unstable forms are radioactive and emit ionizing radiation. Uranium has no stable form and is always radioactive. They are called radionuclides. When a radionuclide decays it transforms into a different atom, a decay product. The parent isotope is the initial product and when it undergoes decay it transforms to form a daughter isotope. They keep on transforming to different atoms until they reach a stable form and no longer radioactive. Half-life is used to measure the decay rate of every nuclide. This is the time required for half of a given number of atoms to decay and it has an inverse proportionality to the isotope's decay constant.

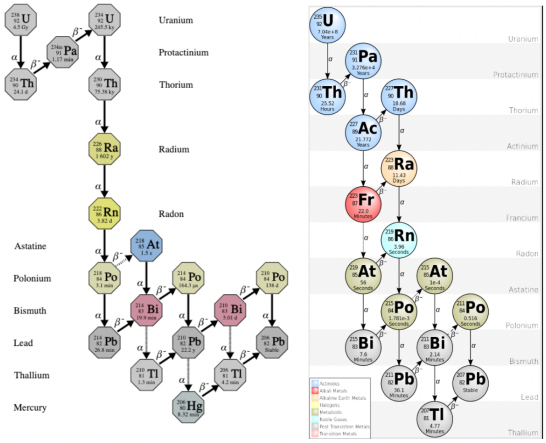


Figure 1: Decay chain for Uranium series(left) and Actinium series(right)

Actinium series:

This is a radioactive series that is initiating with naturally occurring Uranium-235. The chain comprises of unstable heavy atomic nuclei that undergoes decay via a series of alpha, beta decays until a stable nucleus is obtained. In this series the stable nuclei are Lead-207. The alpha decay results in the emission of the nucleus of the helium atom – contains 4 nucleons. Then the mass number will be indicated as 4n+3.

Uranium series:

This radioactive series begins with Uranium-238. The decay chain terminates with Lead-206. Mass number is 4n+2.

Both these series undergo alpha and beta decay. Alpha decay is the disintegration of the nucleus by releasing a Helium atom whilst the beta decay releases a beta particle. Alpha particles contain 2 protons, and 2 neutrons bound together resulting in a heavier mass. As a result, it has higher ionization and short range. Beta particles contain higher energy and greater range of penetration.

In this experiment, the gamma ray spectrum will denote peaks that are corresponding to the gamma rays of the uranium isotopes present.

Gamma ray Spectroscopy –

Study of the energy spectrum of gamma-ray sources. They have a higher photon energy because of its shorter wavelength. As a result, the gamma-ray spectrometer measures the energies of the detected gamma ray photons.

Equations used:

$$number\ of\ atoms\ of\ uranium = \frac{number\ of\ decays\ of\ uranium}{\lambda}$$

$$\lambda = \frac{log_2 2}{t_{1/2}}$$

$$R = \frac{number\ of\ atoms\ of\ U235}{number\ of\ atoms\ of\ U238}$$

The number of atoms can be calculated using the number of gammas in the peak, the efficiency and the relative intensity. λ is the decay constant. The calculation gives us R: the Isotopic abundance and the percentage value of Uranium-235 can be obtained.

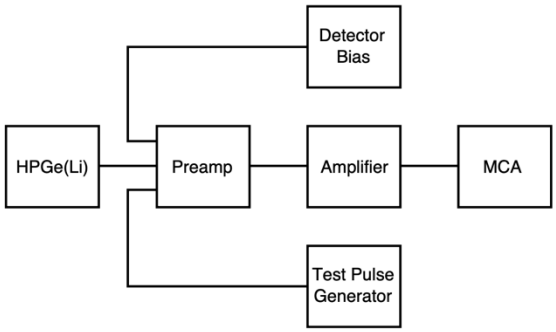
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3. EXPERIMENTAL METHODS:

Components:

The main component used in Gamma ray spectroscopy is the radiation detector that's energy sensitive and the equipment that analyze the output signals. Detectors have the ability to interact with incoming gamma rays due the presence of passive materials. The photoelectric effect, Compton effect and pair production are the most important interaction mechanisms. As a result of these processes, gamma ray energy is absorbed and converted into a voltage signal by analyzing the detector signal before and after the interaction. The signal voltage has a direct proportionality

with the energy of the gamma ray. (Gamma Spectroscopy). Commonly used to detector materials are sodium iodide scintillation counters and the germanium detector. In this experiment we used to the germanium detector.



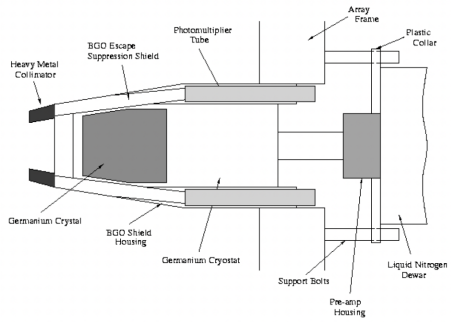


Figure 2: Components of the gamma-ray spectroscopy system *Figure 3: Schematic diagram of Germanium detector*

Germanium Detector: Semiconductor diodes with a p-i-n structure where the intrinsic region is sensitive to ionizing radiation, x-rays and gamma rays. The detector must be cooled as it has a relatively low band gap, and it is required to reduce the thermal generation of charge carriers into an acceptable state.

Preamplifier: Converts the low amplitude current pulses in the detector to voltage pluses.

Amplifier: Increases the size of the pulses through the gain. The gain is used to determine the range of gamma energies visible in the spectrum.

Detector bias: Used in reverse biased semiconductor junction in order to increase the potential difference across the junction.

MCA: It counts the positive square pulses according to their amplitude. Each channel represents a specific voltage.

The fine gain settings are set to 1.195 and course gain settings are set to 20. Therefore, the total gain settings were 23.9 and the shaping period was 3 microseconds.

At the beginning of each day the equipment was stabilized by increasing the voltage by 200V until it reached 1500V. Once the detector was stabilized the next step was to keep the source close to the detector to obtain the energy spectrum. DAQ was used to acquire data. After taking the data for the day we brought down the voltage by 200V until it reached 0.

On the first day we took data on all 5 sources, Cobalt, Cesium, Barium, Uranium and Europium. The data was taken for about 15-20 mins each. This was done to identify the shape of the peaks and the mean positions of the peaks. The second day and third day we took data of Uranium and Europium, and it was taken for a longer period: 30-50 mins each.

The number of counts and the peak positions can be observed in the energy spectra. It was seen that the peek position was relatively same during all three days, but the number of counts would differ. We observed that when the time duration for a particular source is increased the precision is lower which would mean that when the total counts taken per peak is higher it has a $1/\sqrt{N}$ relationship with the precision. In the final two days we observed that the precision fell around the range of 1-2%. Once we obtained the peaks it was required to do background subtraction for noise reduction. Background noise can mask the true values of the peaks therefore by reducing the noise we were able to get a more enhanced clear signal.

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4. ANALYSIS AND RESULTS

Table 1: Data values of the sources used for energy calibration

Element	Bin	Peak	Mean	Rms	Count	EOM	Precision	Gamma
Cesium-137	3558.0	263.0	3556.63	3.70	2302.0	0.08	2.08	661.657
Cobalt - 60	6202.0	145.0	6202.35	5.30	1508.0	0.14	2.58	1173.228
Cobalt - 60	7026.0	120.0	7025.48	5.25	1246.0	0.15	2.83	1332.492
Barium - 133	1563.0	115.0	1563.92	5.00	968.0	0.16	3.21	274.6
Barium - 133	1701.0	257.0	1700.73	3.05	1792.0	0.07	2.36	302.851
Barium - 133	1976.0	669.0	1976.66	2.90	4607.0	0.04	1.47	356.013

Table 2: Data values of Europium

Element	Bin	Peak	Mean	Rms	Count	EOM	Precision	Gamma
Europium – 152								
Day-1	757.0	3485.0	757.29	3.28	19578.0	0.02	0.68	121.0
	1400.0	518.0	1398.0	4.18	2850.0	0.07	1.61	244.7
	1916.0	969.0	1915.57	3.05	6790.5	0.04	1.17	344.3
	4164.0	141.0	4162.26	6.64	1229.0	0.17	2.50	778.9
	5121.0	108.0	5119.43	6.36	1196.0	0.18	2.76	964.08
	5749.0	86.0	5750.62	10.18	794.0	0.32	3.10	1085.87
Day-2	757.0	6069.0	757.37	3.21	33899	0.02	0.51	121.0
	1400.0	887.0	1399.71	3.38	4927.0	0.04	1.26	244.7

	1916.0	1818.0	1915.84	2.97	11880.5	0.03	0.89	344.3
	4164.0	270.0	4163.0	5.04	2122.0	0.10	1.96	778.9
	5123.0	192.0	5120.36	5.87	1953.0	0.12	2.07	964.08
	5751.0	141.0	5750.0	7.25	1345.0	0.18	2.44	1085.87
Day-3	756.0	7743.0	756.50	3.29	43826	0.01	0.45	121.0
	1398.0	1107.0	1398.78	3.51	6292.5	0.04	1.11	244.7
	1914.0	2280.0	1914.61	3.00	15502.5	0.02	0.78	344.3
	4162.0	320.0	4162.38	5.20	2676.0	0.09	1.74	778,9
	5118.0	258.0	5118.63	6.36	2365.5	0.12	1.87	964.08
	5748.0	148.0	5749.37	8.21	1709.5	0.18	2.17	1085.87

Table 3: Data values of Uranium

Element	Bin	Peak	Mean	Rms	Count	EOM	Precision	Gamma	Intensity
Uranium									
Day-1	873.0	1446.0	871.05	9.66	6769.5	0.07	0.71	143.76	10.93
	976.0	915.0	971.57	10.36	3540.0	0.08	0.79	163.33	5.07
	1093.0	7505.0	1092.42	3.50	44885.0	0.02	0.44	185.72	57.2
	1195.0	900.0	1194.78	7.01	5909.5	0.07	0.98	194.94	0.63
	4099.0	261.0	4099.61	15.42	1939.5	0.24	1.59	766.36	0.835
	5313.0	432.0	5311.01	6.22	4538.5	0.09	1.44	1001.7	0.294
Day-2	873.0	3567.0	871.15	9.70	16425	0.04	0.45	143.76	10.93
	976.0	2307.0	971.61	10.48	9109.5	0.05	0.50	163.33	5.07
	1093.0	19068.0	1092.5	3.47	110269.0	0.01	0.28	185.72	57.2
	1195.0	2090.0	1194.86	6.97	14241.5	0.04	0.63	194.94	0.63
	4100.0	672.0	4099.91	15.54	4993.5	0.15	1.00	766.36	0.835
	5313.0	1140.0	5312.38	6.13	11739.5	0.05	0.90	1001.7	0.294
Day-3	872.0	2404.0	870.72	9.67	10813.0	0.05	0.55	143.76	10.93
	975.0	1494.0	971.32	10.42	5682.0	0.06	0.61	163.33	5.07
	1091.0	12223.0	1091.36	3.53	75000.0	0.01	0.34	185.72	57.2
	1194.0	1434.0	1194.40	7.08	9408.0	0.05	0.77	194.94	0.63

4095.0	434.0	4098.02	15.66	3520.0	0.19	1.22	766.36	0.835
5308.0	711.0	5308.12	6.08	7491.5	0.07	1.12	1001.7	0.294

Using the first day data of Cobalt, Cesium and Barium identified the peak positions and the number of counts and the corresponding the gamma energies. This was used to find the energy calibration formula. The graph of the Gamma energy (keV) vs the peak positions was obtained. A linear relationship can be seen. The calibration formula was used to determine the gamma energies for Europium peaks. The accuracy of the formula was found by taking the residual energies. Afterwards the detector efficiencies were found using the number of counts in the peak and the corresponding intensities. Then the Efficiency was plotted against the Europium gamma energy values. This was used to obtain the efficiencies of Uranium. The number of gammas in the peak was obtained using the number of counts and the respective efficiencies of uranium. Then using the calculated number of gamma rays and the branching factor the number of decays in each peak was obtained. Branching factor is a probability of a decay pathway occurring in comparison to other possible pathways in a decay chain. Then the number of atoms in each peak can be obtained by taking the number of gammas and the decay constant for that isotope. Then finally the isotopic abundance ratio was calculated by taking the ratio of the number of atoms in Uranium-235 and Uranium-238. Identification of the sample Uranium-235, as if its natural, enriched or depleted is by getting the percentage value.

Energy calibration:

$y = 0.19335x^2 - 26.01512$

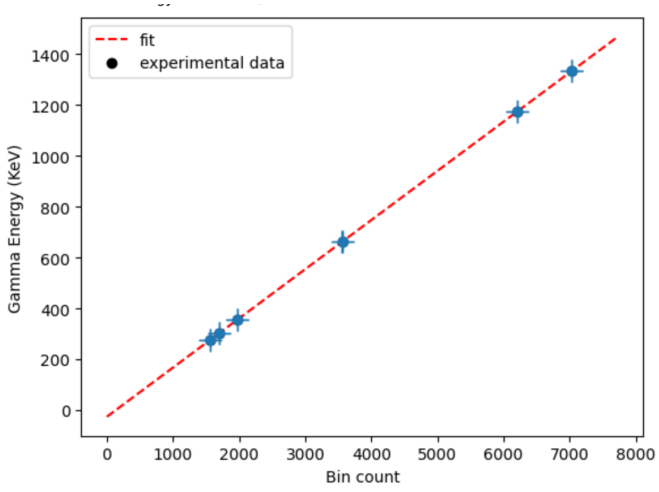


Figure 4: Energy calibration plot

Table 4: The residual energy comparison with Europium data

Literature value	*Experimental – Day1*	*Residual – Day 1*	*Experimental – Day 2*	*Residual – Day 2*	*Experimental – Day 3*	*Residual – Day 3*
121.0	120.35	-0.6490	120.35	-0.6490	120.16	-0.8423
244.7	244.68	-0.0248	244.68	-0.0248	244.29	-0.4115
344.3	344.44	0.1438	344.44	0.1438	344.06	-0.2428
778.9	779.09	0.1951	779.09	0.1951	778.71	-0.1915
964.08	964.13	0.0523	964.52	0.4390	963.55	-0.5277
1085.87	1085.55	-0.3137	1085.94	0.0729	1085.36	-0.5071

The residual energies are extremely small which would mean that the energy fit function is accurate. Data values on day 1 and day 2 are very similar and minor variation can be observed in day 3.

Relative Detector Efficiency:

In order to calculate the relative detector efficiency a reference peak needed to be chosen in the Europium data spectrum. Here we considered the peak position that's corresponding the gamma ray 121.00 keV. The number of counts and the intensity are corresponding to that particular peak. We assume that the reference peak has 100% efficiency with respect to itself.

Table 5: Reference values used for relative detector efficiency calculation

	N_ref	σN_ref	I_ref	σI_ref
Day 1	19578.1	139.8	28.37	0.13
Day 2	33899.0	183.9	28.37	0.13
Day 3	43826.0	209.4	28.37	0.13

Day 1		Day 2		Day 3:			
N_i	σN_i	N_i	σN_i	N_i	σN_i	I_i	σI_i
2850.0	53.39	4927.0	70.19	6292.5	79.32	7.53	0.04
6790.5	82.40	11880.0	108.99	15502.5	124.50	26.57	0.11
1229.0	35.05	2122.0	46.06	2676.0	51.73	12.97	0.06
1196.0	34.58	1953.0	44.19	2356.5	48.54	14.63	0.06
794.5	28.18	1345.0	36.67	1709.5	41.34	10.13	0.05

Table 6: Efficiency values of Europium

	Day 1:		Day 2:		Day 3:	
Gamma energy	Efficiency	σEff	Efficiency	σEff	Efficiency	σEff
143.76	0.5485	0.0150	0.5475	0.0086	0.5409	0.0101
163.33	0.3703	0.0067	0.3742	0.0041	0.3777	0.0218
185.715	0.1373	0.0056	0.1369	0.0031	0.1336	0.0038
766.36	0.1184	0.0048	0.1117	0.0026	0.1043	0.0030

1001.7	0.1136	0.0057	0.1111	0.0031	0.1092	0.0051

Day 1:

$$y = 8.766 * 10^{-6} * x^2 - 0.001635 * x + 0.8678$$

Day 2:

$$y = 8.671 * 10^{-7} * x^2 - 0.001629 * x + 0.8682$$

Day 3:

$$y = 8.603 * 10^{-7} * x^2 - 0.001623 * x + 0.864$$

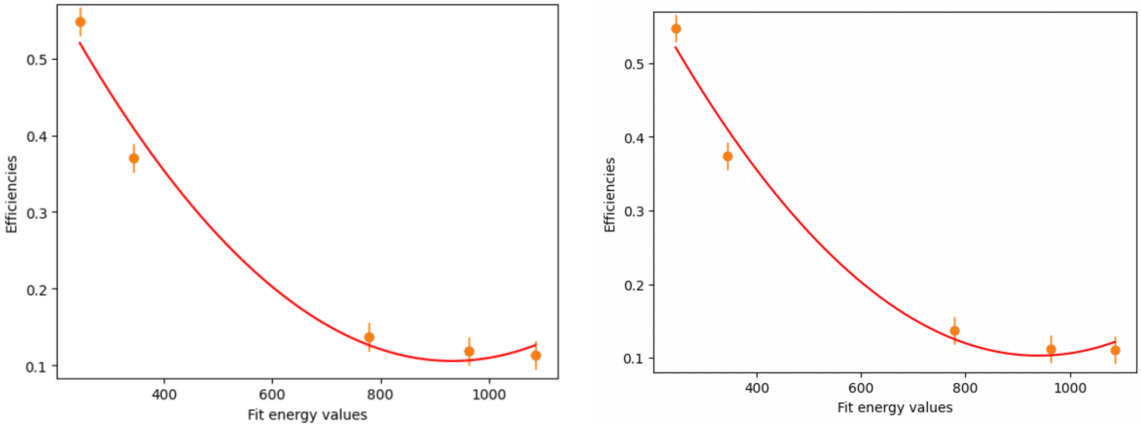


Figure 5: Day 1 and Day 2

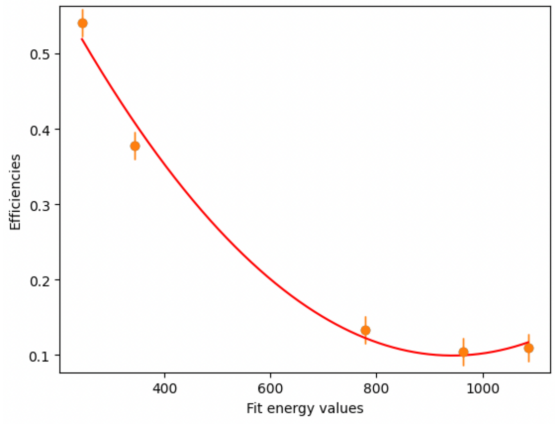


Figure 6: Day 3

Isotopic Abundance calculation:

Decay constants:

$$\lambda_{235} = 9.8542 * 10^{-1}$$

$$\lambda_{238} = 1.55101 * 10^{-1}$$

Table 7: Efficiencies and the Isotopic ratio U235/U238

	Gamma energies (keV)	Number of counts	Efficiencies	Branching ratios	Decay counts	Number of atoms	Isotopic ratio
Day 1:							0.00294967 +/- 0.00007596
U-235	143.76	6769.5	0.6509	57.2 +/- 0.9	951.527	0.66696e12	
U-238	766.36	1939.5	0.1298	0.842 +/- 0.008	50814.07	2.2270e14	
Day 2:							0.00274723 +/- 0.00004439
U-235	143.76	16425.0	0.6519	57.2 +/- 0.9	2298.77	1.6178e12	
U-238	766.36	4993.5	0.1289	0.842 +/- 0.008	131806.4	5.8891e14	
Day 3:							0.00252225 +/- 0.00004894
U-235	143.76	10813	0.6492	57.2 +/- 0.9	1519.78	1.0696e12	
U-238	766.36	3520	0.1261	0.842 +/- 0.008	94913.86	4.2407e14	

The efficiencies of Europium are very similar to one another. As the gamma energies increase the efficiencies decrease. Therefore, the fit function is a decreasing second order polynomial function. The half-life of Uranium-235 and Uranium-238 are used to calculate the decay constants. Efficiency values of Uranium show a very slight variation within the three days. Here for Uranium-235 the considered gamma energy value is 143.76 keV and for Uranium-238 the considered gamma energy value is 766.36 keV. The isotopic ratios are in the range of 0.002 and it stays consistent along the three days. Therefore, it can be concluded that the uranium sample is depleted as its below 0.007.

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5. CONCLUSION:

Calibration of the detector was done using Cobalt, Barium and Cesium. This was used to obtain the gamma energies of Europium. The residual energies were below 1 keV. Detector efficiency and the isotopic ratio was calculated using the obtained data. It can be said that the relative efficiency of Uranium-235 is higher than Uranium-238. It was observed that the isotopic ratio for Uranium-235 is 0.002 which would mean that the uranium sample used is depleted.

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6. REFERENCES

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